

RSVP PROJECT – MYSQL

“SQL Movie Database Analysis”

QUESTIONS AND QUERIES BELOW WITH
SCREENSHOT OF OUTPUTS

Q1. Find the total number of rows in each table of the schema?

QUERY-

```
SELECT COUNT(*) AS movie_row_count
FROM movie;
```

```
SELECT COUNT(*) AS genre_row_count
FROM genre;
```

```
SELECT COUNT(*) AS names_row_count
FROM names;
```

```
SELECT COUNT(*) AS ratings_row_count
FROM ratings;
```

```
SELECT COUNT(*) AS director_mapping_row_count
FROM director_mapping;
```

```
SELECT COUNT(*) AS role_mapping_row_count
FROM role_mapping;
```

OUTPUT

	movie_row_count
▶	7997
	genre_row_count
▶	14662
	names_row_count
▶	25735
	ratings_row_count
▶	7997
	director_mapping_row_count
▶	3867
	role_mapping_row_count
▶	15615

Q2. Which columns in the 'movie' table have null values?

QUERY-

```
SELECT COUNT(*) AS title_nulls
FROM movie
WHERE title IS NULL;
```

```
SELECT COUNT(*) AS year_nulls
FROM movie
WHERE year IS NULL;
```

```
SELECT COUNT(*) AS worldwide_gross_income_nulls
FROM movie
WHERE worldwide_gross_income IS NULL;
```

```
SELECT COUNT(*) AS date_published_nulls
FROM movie
WHERE date_published IS NULL;
```

```
SELECT COUNT(*) AS duration_nulls
FROM movie
WHERE duration IS NULL;
```

```
SELECT COUNT(*) AS country_nulls
FROM movie
WHERE country IS NULL;
```

```
SELECT COUNT(*) AS languages_nulls
FROM movie
WHERE languages IS NULL;
```

```
SELECT COUNT(*) AS production_company_nulls
FROM movie
WHERE production_company IS NULL;
```

OUTPUT

title_nulls	0
year_nulls	0
worldwide_gross_income_nulls	3724
date_published_nulls	0
duration_nulls	0
country_nulls	20
languages_nulls	194
production_company_nulls	528

Q3.1 Find the total number of movies released in each year.

/* Output format :

Year	number_of_movies
2017	2134
2018	-
2019	-

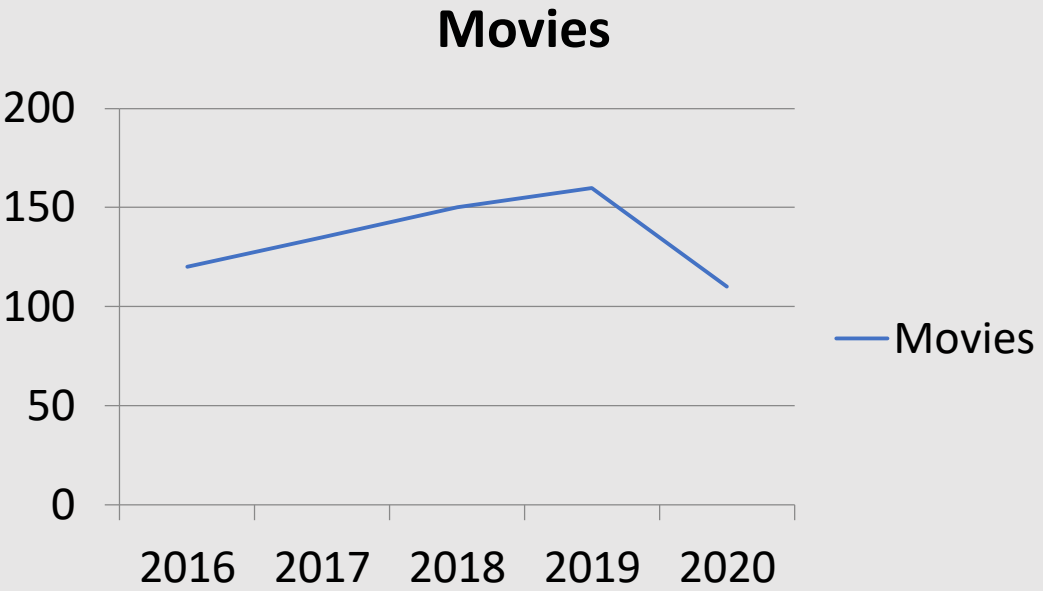
*/

QUERY-

```
SELECT year, COUNT(*) AS number_of_movies
FROM movie
GROUP BY year;
```

OUTPUT

	year	number_of_movies
▶	2017	3052
	2018	2944
	2019	2001



Q3.2 How does the trend look month-wise? (Output expected)

```
/* Output format :
+-----+-----+
| month_num | number_of_movies |
+-----+-----+
|      1      |      134      |
|      2      |      231      |
|      3      |      -        |
+-----+-----+ */
```

QUERY-

```
SELECT MONTH(date_published) AS month_num,
        COUNT(*) AS number_of_movies
FROM movie
GROUP BY month_num
ORDER BY month_num ;
```

OUTPUT

	month_num	number_of_movies
▶	1	804
	2	640
	3	824
	4	680
	5	625
	6	580
	7	493
	8	678
	9	809
	10	801
	11	625

Q4. How many movies were produced in the USA or India in the year 2019?

QUERY- **SELECT** COUNT(*) **AS** number_of_movies

FROM movie

WHERE country **LIKE** '%USA%'
OR country **LIKE** '%INDIA%'
AND year **LIKE** '%2019%';

OUTPUT

	number_of_movies
▶	3164

Q5. Find the unique list of the genres present in the data set?

QUERY- **SELECT distinct** genre **AS** genre

FROM genre

ORDER BY genre;

OUTPUT

	genre
▶	Action
	Adventure
	Comedy
	Crime
	Drama
	Family
	Fantasy
	Horror
	Mystery
	Others
	Romance
	Sci-Fi

Q6.Which genre had the highest number of movies produced overall?

/* Output format :

genre	movie_count
-	-
-	-
-	-

*/

QUERY-

```
SELECT genre,
        COUNT(*) AS movie_count
FROM genre AS g JOIN
movie AS m ON g.movie_id = m.id

GROUP BY
        genre
ORDER BY
        movie_count desc
LIMIT 1,
```

OUTPUT

	genre	movie_count
►	Drama	4285

Q7. How many movies belong to only one genre?

QUERY-

```
WITH movie_genre_summary AS (  
  SELECT  g.movie_id,  
          COUNT(*) AS genre_count  
  FROM    movie AS m JOIN  
  genre AS g ON m.id = g.movie_id  
  GROUP BY g.movie_id)  
SELECT COUNT(*) AS  
single_genre_movie_count  
FROM movie_genre_summary  
WHERE genre_count = 1;
```

OUTPUT



	single_genre_movie_count
▶	3289

Q8.What is the average duration of movies in each genre?

QUERY-

SELECT

g.genre, **AVG**(m.duration) **AS** avg_duration

FROM

genre **AS** g

LEFT JOIN

movie **AS** m **ON** g.movie_id = m.id

GROUP BY

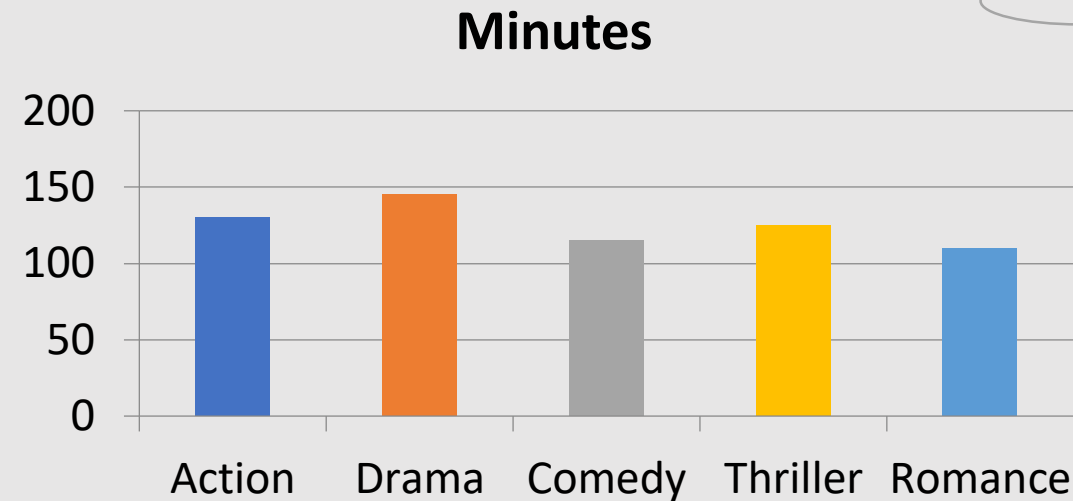
genre

ORDER BY

avg_duration **desc**

OUTPUT

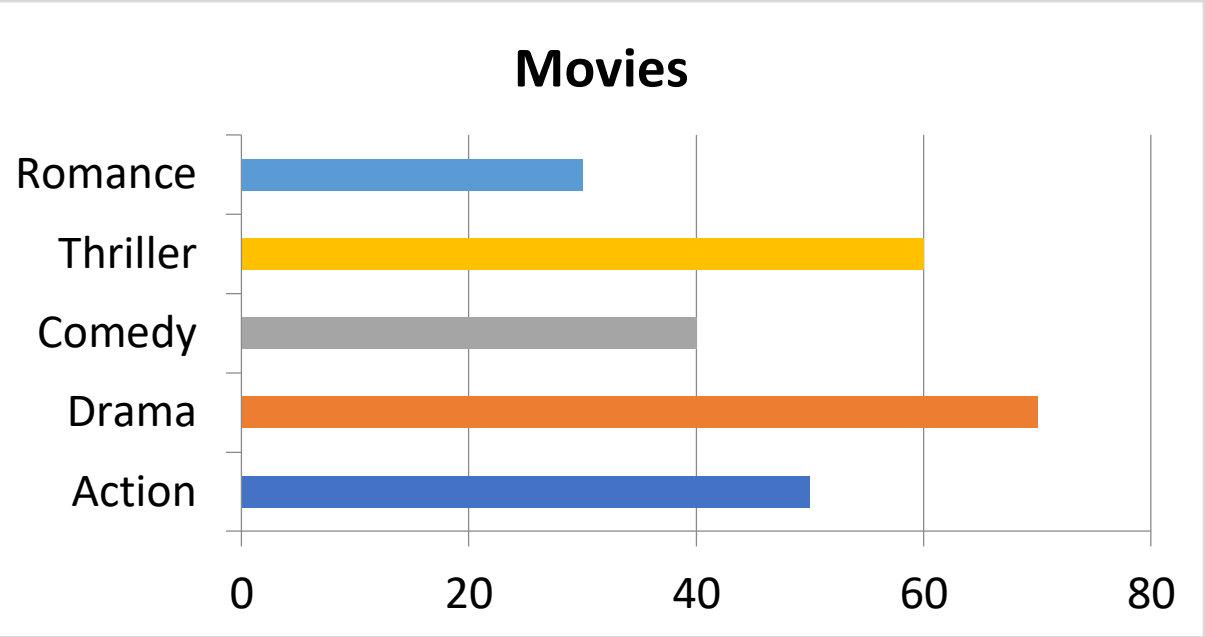
	genre	avg_duration
▶	Action	112.8829
	Romance	109.5342
	Crime	107.0517
	Drama	106.7746
	Fantasy	105.1404
	Comedy	102.6227
	Adventure	101.8714
	Mystery	101.8000
	Thriller	101.5761
	Family	100.9669
	Others	100.1600
	Sci-Fi	97.9413



Q9.What is the rank of the 'thriller' genre of movies among all the genres in terms of number of movies produced?

/* Output format:

genre	movie_count	genre_rank
-	-	-



QUERY-

```
WITH summary AS
(
  SELECT
    genre,
    COUNT(movie_id) AS movie_count
    RANK() OVER (ORDER BY COUNT(movie_id) DESC) AS genre_rank
  FROM
    genre
  GROUP BY genre )
SELECT COUNT(*)
FROM summary
WHERE genre = 'Thriller';
```

OUTPUT

	genre	movie_count	genre_rank
▶	Thriller	1484	3

Q10. Find the minimum and maximum values for each column of the 'ratings' table except the movie_id column.

/* Output format:

min_avg_rating	max_avg_rating	min_total_votes	max_total_votes	min_median_rating	max_median_rating
0	5	177	2000	0	8

QUERY-

SELECT

MIN(avg_rating) **AS** min_avg_rating,
MAX(avg_rating) **AS** max_avg_rating,
MIN(total_votes) **AS** min_total_votes,
MAX(total_votes) **AS** max_total_votes,
MIN(median_rating) **AS** min_median_rating,
MAX(median_rating) **AS** max_median_rating

FROM ratings;

OUTPUT

	min_avg_rating	max_avg_rating	min_total_votes	max_total_votes	min_median_rating	max_median_rating
▶	1.0	10.0	100	725138	1	10

Q11. What are the top 10 movies based on average rating?

QUERY-

```
WITH top_movies AS (  
  SELECT m.title, r.avg_rating,  
         ROW_NUMBER () OVER (ORDER BY r.avg_rating DESC) AS movie_rank  
  FROM  
    movie AS m  
    INNER JOIN  
    ratings AS r ON m.id = r.movie_id )  
SELECT *  
FROM top_movies  
LIMIT 10;
```

OUTPUT

	title	avg_rating	movie_rank
▶	Kirket	10.0	1
	Love in Kilnerry	10.0	2
	Gini Helida Kathe	9.8	3
	Runam	9.7	4
	Fan	9.6	5
	Android Kunjappan Version 5.25	9.6	6
	Yeh Suhaagraat Impossible	9.5	7
	Safe	9.5	8
	The Brighton Miracle	9.5	9
	Shibu	9.4	10

Q12. Summarise the ratings table based on the movie counts by median ratings.(order by median_rating)

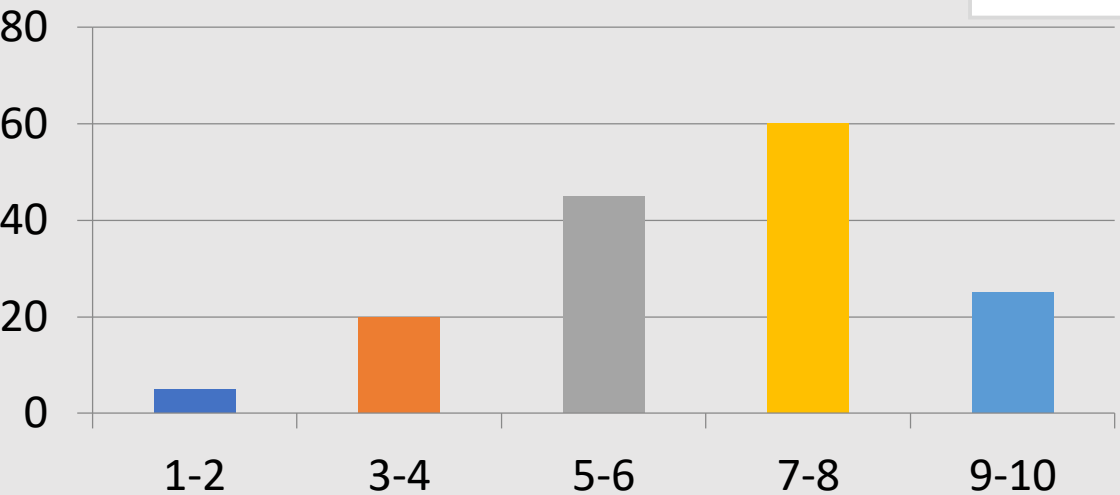
QUERY-

```
SELECT
    median_rating,
    COUNT(*) AS movie_count
FROM
    ratings
GROUP BY
    median_rating
ORDER BY
    movie_count DESC;
```

OUTPUT

	median_rating	movie_count
▶	7	2257
	6	1975
	8	1030
	5	985
	4	479
	9	429
	10	346
	3	283
	2	119
	1	94

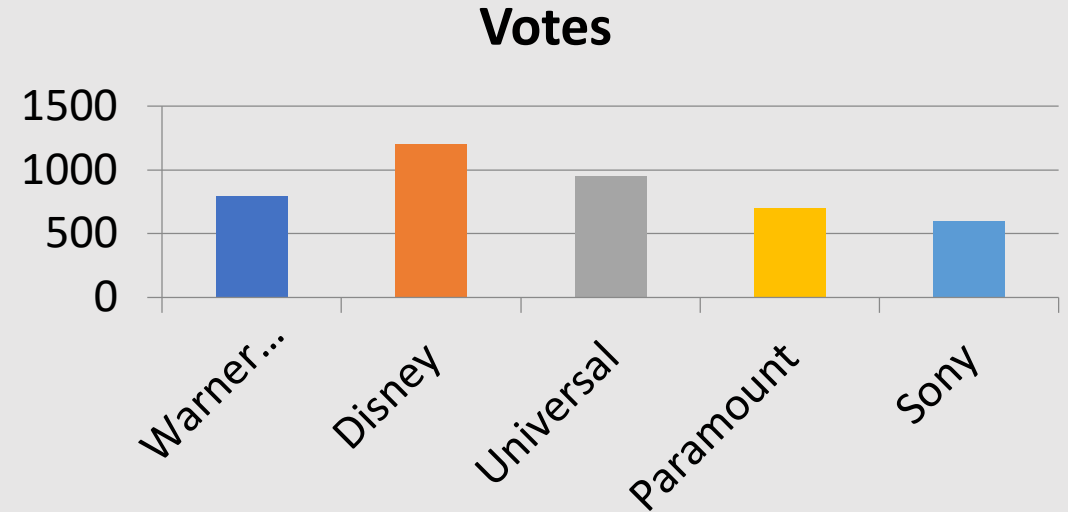
Movies



Q13. Which production house has produced the most number of hit movies (average rating > 8)?

QUERY-

```
WITH top_prod AS (  
  SELECT m.production_company,  
         COUNT(m.id) AS movie_count,  
         RANK() OVER (ORDER BY COUNT(m.id) DESC)  
         AS prod_company_rank  
  FROM movie AS m  
       INNER JOIN  
       ratings AS r  
       ON m.id = r.movie_id  
 WHERE m.production_company IS NOT NULL AND r.avg_rating > 8  
 GROUP BY m.production_company)  
 SELECT *  
 FROM top_prod  
 LIMIT 1;
```



OUTPUT

	production_company	movie_count	prod_company_rank
▶	Dream Warrior Pictures	3	1

Q14. How many movies released in each genre in March 2017 in the USA had more than 1,000 votes?(Split the question into parts and try to understand it.)

QUERY- **SELECT** g.genre,
COUNT(g.movie_id) **AS** movie_count
FROM genre **AS** g
INNER JOIN
movie **AS** m
ON g.movie_id = m.id
INNER JOIN
ratings **AS** r
ON m.id = r.movie_id
WHERE m.year = 2017 **AND** MONTH(m.date_published) = 3 **AND**
m.country **LIKE** '%USA%' **AND** r.total_votes > 1000
GROUP BY
g.genre
ORDER BY
movie_count **DESC**;

OUTPUT

	genre	movie_count
►	Drama	24
	Comedy	9
	Action	8
	Thriller	8
	Sci-Fi	7
	Crime	6
	Horror	6
	Mystery	4
	Romance	4
	Fantasy	3
	Adventure	3
	Family	1

Q15. Find the movies in each genre that start with the characters 'The' and have an average rating > 8.

QUERY-

```
SELECT  m.title,
          r.avg_rating,
          g.genre
FROM    movie AS m
INNER JOIN
          rating AS r
ON       m.id = r.movie_id
INNER JOIN
          genre AS g
ON       m.id = g.movie_id
WHERE
          m.title LIKE 'The%' AND r.avg_rating > 8
ORDER BY
          r.avg_rating DESC;
```

OUTPUT

	title	avg_rating	genre
▶	The Brighton Miracle	9.5	Drama
	The Colour of Darkness	9.1	Drama
	The Blue Elephant 2	8.8	Drama
	The Blue Elephant 2	8.8	Horror
	The Blue Elephant 2	8.8	Mystery
	The Irishman	8.7	Crime
	The Irishman	8.7	Drama
	The Mystery of Godliness: The Se...	8.5	Drama
	The Gambinos	8.4	Crime
	The Gambinos	8.4	Drama
	Theeran Adhigaaram Ondru	8.3	Action
	Theeran Adhigaaram Ondru	8.3	Crime

Q16. Of the movies released between 1 April 2018 and 1 April 2019, how many were given a median rating of 8?

QUERY-

SELECT

COUNT(*) **AS** movie_count

FROM movie **AS** m

INNER JOIN

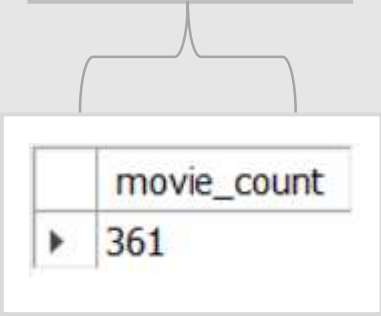
ratings **AS** r

ON m.id = r.movie_id

WHERE m.date_published **BETWEEN** '2018-04-01' **AND** '2019-04-01' **AND**

r.median_rating = 8;

OUTPUT



	movie_count
▶	361

Q17. Do German movies get more votes than Italian movies?

QUERY-

```
WITH language_votes AS(
SELECT
    SUM(CASE WHEN LOWER(languages) LIKE '%german%' THEN total_votes END) AS german_votes,
    COUNT(CASE WHEN LOWER(languages) LIKE '%german%' THEN movie_id END) AS german_movies,
    SUM(CASE WHEN LOWER(languages) LIKE '%italian%' THEN total_votes END) AS italian_votes,
    COUNT(CASE WHEN LOWER(languages) LIKE '%italian%' THEN movie_id END) AS italian_movies
FROM movie AS m
INNER JOIN
    ratings AS r
    ON m.id = r.movie_id)
SELECT ROUND(german_votes / german_movies, 2) AS german_votes_per_movie,
    ROUND(italian_votes / italian_movies, 2) AS italian_votes_per_movie
FROM language_votes;
```

OUTPUT

	german_votes_per_movie	italian_votes_per_movie
▶	12928.44	11960.47

Q18. Find the number of null values in each column of the 'names' table, except for the 'id' column.

QUERY-

SELECT

COUNT(CASE WHEN name IS NULL THEN id END) AS name_nulls,
COUNT(CASE WHEN height IS NULL THEN id END) AS height_nulls,
COUNT(CASE WHEN date_of_birth IS NULL THEN id END) AS date_of_birth_nulls,
COUNT(CASE WHEN known_for_movies IS NULL THEN id END) AS known_for_movies_nulls

FROM

names;

OUTPUT

	name_nulls	height_nulls	date_of_birth_nulls	known_for_movies_nulls
▶	0	17335	13431	15226

Q19. Who are the top three directors in each of the top three genres whose movies have an average rating > 8?

QUERY-

```
WITH topRatedGenres AS(
SELECT  genre, COUNT(m.id) AS movie_count,
        RANK () OVER (ORDER BY COUNT(m.id) DESC) AS genre_rank
FROM    genre AS g
        LEFT JOIN movie AS m ON g.movie_id = m.id
        INNER JOIN ratings AS r ON m.id=r.movie_id where avg_rating>8
GROUP BY genre)
SELECT  n.name AS director_name, COUNT(m.id) AS movie_count
FROM    names AS n
        INNER JOIN director_mapping AS d ON n.id=d.name_id
        INNER JOIN movie AS m ON d.movie_id = m.id
        INNER JOIN ratings AS r ON m.id=r.movie_id
        INNER JOIN genre AS g ON g.movie_id = m.id
WHERE   g.genre IN (
        SELECT DISTINCT genre
        FROM topRatedGenres
        WHERE genre_rank<=3) AND avg_rating>8
GROUP BY name
ORDER BY movie_count DESC
LIMIT 3;
```

OUTPUT

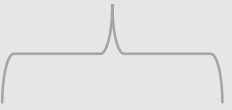
	director_name	movie_count
▶	James Mangold	4
	Joe Russo	3
	Anthony Russo	3

Q20. Who are the top two actors whose movies have a median rating ≥ 8 ?

QUERY-

```
SELECT n.name as actor_name,
       COUNT(m.id) AS movie_count
FROM   name AS n
       INNER JOIN role_mapping AS rm ON n.id = rm.name_id
       INNER JOIN movie AS m ON rm.movie_id = m.id
       INNER JOIN ratings AS r ON m.id = r.movie_id
WHERE  rm.category = 'actor' AND r.median_rating  $\geq$  8
GROUP BY
       n.name
ORDER BY
       movie_count DESC
LIMIT 2;
```

OUTPUT



	actor_name	movie_count
▶	Mammootty	8
	Mohanlal	5

Q21. Which are the top three production houses based on the number of votes received by their movies?

QUERY-

```
WITH top_prod AS(
SELECT m.production_company, SUM(r.total_votes) AS vote_count,
       ROW_NUMBER() OVER(ORDER BY SUM(r.total_votes) DESC)
       AS prod_comp_rank
FROM   movie AS m LEFT JOIN ratings as r ON m.id = r.movie_id
WHERE  m.production_company IS NOT NULL
GROUP BY m.production_company)
SELECT *
FROM top_prod
LIMIT 3;
```

OUTPUT



	production_company	vote_count	prod_comp_rank
▶	Marvel Studios	2656967	1
	Twentieth Century Fox	2411163	2
	Warner Bros.	2396057	3

Q22. Rank actors with movies released in India based on their average ratings. Which actor is at the top of the list?

QUERY-

```
WITH actor_ratings AS(
SELECT  n.name AS actor_name,
        SUM(r.total_votes) AS total_votes,
        COUNT(m.id) as movie_count,
        ROUND(
            SUM(r.avg_rating*r.total_votes) /SUM(r.total_votes), 2)
        AS actor_avg_rating
FROM  name AS n
      INNER JOIN role_mapping AS rm ON n.id=rm.name_id
      INNER JOIN movies AS m ON rm.movie_id = m.id
      INNER JOIN rating AS r ON m.id=r.movie_id
WHERE  category = 'actor' AND LOWER(country)
      LIKE '%INDIA%'
GROUP BY actor_name )
SELECT *,
        RANK() OVER (ORDER BY actor_avg_rating
                      DESC, total_votes DESC) AS actor_rank
FROM actor_ratings
WHERE movie_count>= 5;
```

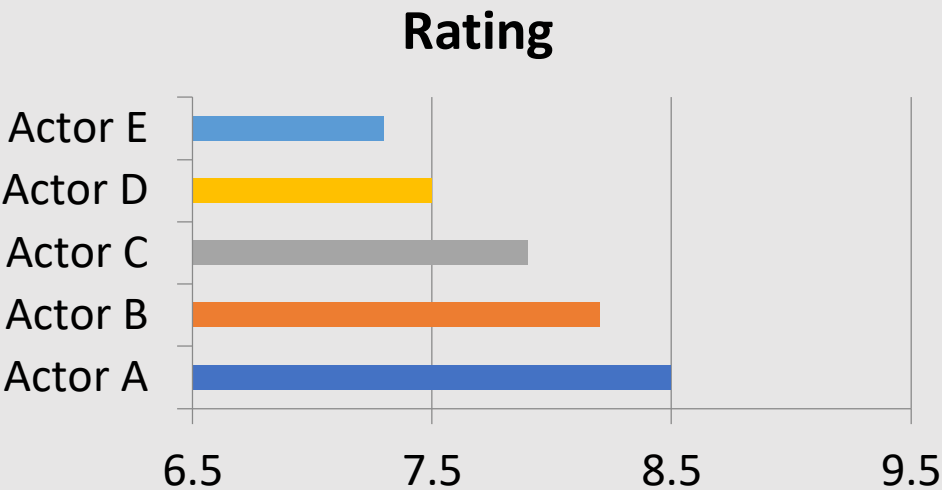
OUTPUT

	actor_name	total_votes	movie_count	actor_avg_rating	actor_rank
▶	Vijay Sethupathi	23114	5	8.42	1
	Fahadh Faasil	13557	5	7.99	2
	Yogi Babu	8500	11	7.83	3
	Joju George	3926	5	7.58	4
	Ammy Virk	2504	6	7.55	5
	Dileesh Pothan	6235	5	7.52	6
	Kunchacko Boban	5628	6	7.48	7
	Pankaj Tripathi	40728	5	7.44	8
	Rajkummar Rao	42560	6	7.37	9
	Dulquer Salmaan	17666	5	7.30	10
	Amit Sadh	13355	5	7.21	11
	Tovino Thomas	11596	8	7.15	12

Q23.Find the top five actresses in Hindi movies released in India based on their average ratings.

QUERY-

```
WITH actress_ratings AS(
SELECT  n.name AS actress_name,
        SUM(r.total_votes) AS total_votes,
        COUNT(m.id) AS movie_count,
        ROUND(SUM(r.avg_rating*r.total_votes)/SUM(r.total_votes) ,2)
        AS actress_avg_rating
FROM  names AS n INNER JOIN
      role_mapping AS rm
      ON n.id=rm.name_id INNER JOIN
      movie AS m   ON rm.movie_id = m.id
      INNER JOIN ratings AS r  ON m.id=r.movie_id
WHERE rm.category = 'actress' AND
      LOWER(m.languages) LIKE '%hindi%' AND
      LOWER(m.country) LIKE '%india%'
GROUP BY actress_name)
SELECT *,
        RANK() OVER (ORDER BY actress_avg_rating DESC,
                      total_votes DESC) AS actress_rank
FROM  actress_ratings
WHERE movie_count>=3
LIMIT 5;
```



OUTPUT

	actress_name	total_votes	movie_count	actress_avg_rating	actress_rank
▶	Taapsee Pannu	18061	3	7.74	1
	Kriti Sanon	21967	3	7.05	2
	Divya Dutta	8579	3	6.88	3
	Shraddha Kapoor	26779	3	6.63	4
	Kriti Kharbanda	2549	3	4.80	5

Q24. Consider thriller movies having at least 25,000 votes. Classify them according to their average ratings in the following categories:

- Rating > 8: Superhit
- Rating between 7 and 8: Hit
- Rating between 5 and 7: One-time-watch
- Rating < 5: Flop */

```
QUERY- SELECT m.title AS movie_name,
        CASE
        WHEN r.avg_rating > 8 THEN 'Superhit'
        WHEN r.avg_rating BETWEEN 7 AND 8 THEN 'Hit'
        WHEN r.avg_rating BETWEEN 5 AND 7 THEN 'One-time-watch'
        ELSE 'Flop'
        END AS movie_category
FROM movie AS m
INNER JOIN ratings AS r ON m.id = r.movie_id
INNER JOIN genre AS g ON m.id = g.movie_id
WHERE g.genre = 'Thriller' AND r.total_votes > 25000
ORDER BY r.avg_rating DESC;
```

OUTPUT

	movie_name	movie_category
▶	Joker	Superhit
	Andhadhun	Superhit
	Ah-ga-ssi	Superhit
	Contratiempo	Superhit
	Mission: Impossible - Fallout	Hit
	Forushande	Hit
	Get Out	Hit
	Searching	Hit
	Hotel Mumbai	Hit
	John Wick: Chapter 3 - Parabellum	Hit
	John Wick: Chapter 2	Hit
	Miss Sloane	Hit

Q25. What is the genre-wise running total and moving average of the average movie duration?

```
QUERY- WITH genre_summary AS(
SELECT g.genre,
      ROUND(AVG(m.duration), 2) AS avg_duration
FROM genre AS g
      LEFT JOIN movie AS m
      ON g.movie_id = m.id
GROUP BY g.genre)
SELECT genre, avg_duration,
      SUM(avg_duration) OVER (ORDER BY genre)
      AS running_total_duration,
      AVG(avg_duration) OVER (ORDER BY genre
      ROWS BETWEEN UNBOUNDED PRECEDING AND
      CURRENT ROW) AS moving_avg_duration
FROM genre_summary;
```

OUTPUT

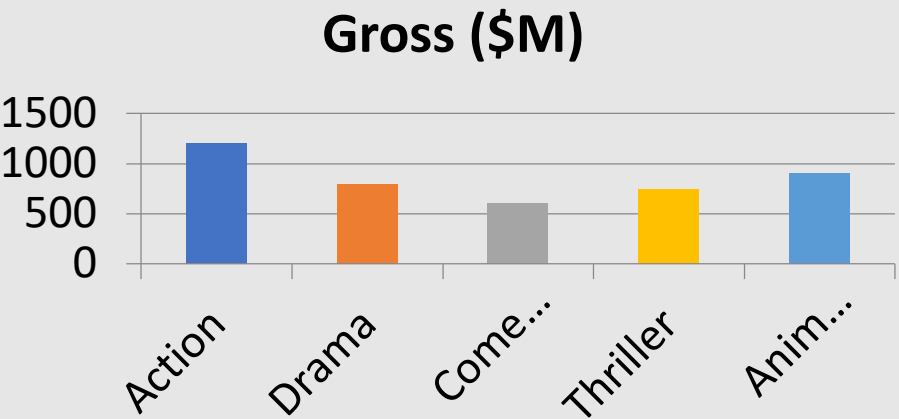
	genre	avg_duration	running_total_duration	moving_avg_duration
▶	Action	112.88	112.88	112.880000
	Adventure	101.87	214.75	107.375000
	Comedy	102.62	317.37	105.790000
	Crime	107.05	424.42	106.105000
	Drama	106.77	531.19	106.238000
	Family	100.97	632.16	105.360000
	Fantasy	105.14	737.30	105.328571
	Horror	92.72	830.02	103.752500
	Mystery	101.80	931.82	103.535556
	Others	100.16	1031.98	103.198000
	Romance	109.53	1141.51	103.773636
	Sci-Fi	97.94	1239.45	103.287500

Q26. Which are the five highest-grossing movies in each year for each of the top three genres?

QUERY-

```
WITH top_genres AS (
SELECT genre,
COUNT(m.id) AS movie_count,
RANK () OVER (ORDER BY COUNT(m.id) DESC) AS genre_rank
FROM genre AS g LEFT JOIN movie AS m
ON g.movie_id = m.id
GROUP BY genre),
top_grossing AS(
SELECT g.genre, year, m.title as movie_name, worldwide_gross_income,
RANK() OVER (PARTITION BY g.genre, year
ORDER BY CONVERT(REPLACE(TRIM(worldwide_gross_income), "$ ", ""),
UNSIGNED INT) DESC) AS movie_rank
FROM movie AS m INNER JOIN genre AS g ON g.movie_id = m.id
WHERE g.genre IN (SELECT DISTINCT genre
FROM top_genres
WHERE genre_rank<=3))

SELECT *
FROM top_grossing
WHERE movie_rank<=5;
```



OUTPUT

	genre	year	movie_name	worldwide_gross_income	movie_rank
▶	Comedy	2017	Despicable Me 3	\$ 1034799409	1
	Comedy	2017	Jumanji: Welcome to the Jungle	\$ 962102237	2
	Comedy	2017	Guardians of the Galaxy Vol. 2	\$ 863756051	3
	Comedy	2017	Thor: Ragnarok	\$ 853977126	4
	Comedy	2017	Sing	\$ 634151679	5
	Comedy	2018	Deadpool 2	\$ 785046920	1
	Comedy	2018	Ant-Man and the Wasp	\$ 622674139	2
	Comedy	2018	Tang ren jie tan an 2	\$ 544061916	3
	Comedy	2018	Ralph Breaks the Internet	\$ 529323962	4
	Comedy	2018	Hotel Transylvania 3: Summer Va...	\$ 528583774	5
	Comedy	2019	Toy Story 4	\$ 1073168585	1
	Comedy	2019	Pokémon Detective Pikachu	\$ 431705346	2

Q27. What are the top two production houses that have produced the highest number of hits (median rating ≥ 8) among multilingual movies?

QUERY-

```
WITH top_prod AS(
SELECT m.production_company,
      COUNT(m.id) AS movie_count,
      RANK() OVER (ORDER BY COUNT(m.id) DESC) AS prod_comp_rank
FROM movie AS m
      INNER JOIN
      ratings AS r ON m.id = r.movie_id
WHERE r.median_rating >= 8 AND
      m.production_company IS NOT NULL AND POSITION(',') IN m.languages) > 0
GROUP BY m.production_company)
SELECT *
FROM top_prod
LIMIT 2;
```

OUTPUT

	production_company	movie_count	prod_comp_rank
▶	Star Cinema	7	1
	Twentieth Century Fox	4	2

Q27. What are the top two production houses that have produced the highest number of hits (median rating ≥ 8) among multilingual movies?

QUERY-

```
WITH top_prod AS(
SELECT m.production_company,
      COUNT(m.id) AS movie_count,
      RANK() OVER (ORDER BY COUNT(m.id) DESC) AS prod_comp_rank
FROM movie AS m
      INNER JOIN
      ratings AS r ON m.id = r.movie_id
WHERE r.median_rating >= 8 AND
      m.production_company IS NOT NULL AND POSITION(',') IN m.languages) > 0
GROUP BY m.production_company)
SELECT *
FROM top_prod
LIMIT 2;
```

OUTPUT

	production_company	movie_count	prod_comp_rank
▶	Star Cinema	7	1
	Twentieth Century Fox	4	2

Q28. Who are the top 3 actresses based on the number of Super Hit movies (average rating > 8) in 'drama' genre?

QUERY-

```
WITH actress_ratings AS(
SELECT n.name as actress_name,
      SUM(r.total_votes) AS total_votes,
      COUNT(m.id) as movie_count,
      ROUND(SUM(r.avg_rating*r.total_votes)/SUM(r.total_votes),2) AS
      actress_avg_rating
FROM names AS n
      INNER JOIN role_mapping AS a ON n.id=a.name_id
      INNER JOIN movie AS m ON a.movie_id = m.id
      INNER JOIN ratings AS r ON m.id=r.movie_id
      INNER JOIN genre AS g ON m.id=g.movie_id
WHERE category = 'actress' AND lower(g.genre) = 'drama'
GROUP BY actress_name)
SELECT *,
      ROW_NUMBER() OVER (ORDER BY actress_avg_rating DESC,
      total_votes DESC) AS actress_rank
FROM actress_ratings
LIMIT 3;
```

OUTPUT

	actress_name	total_votes	movie_count	actress_avg_rating	actress_rank
▶	Sangeetha Bhat	1010	1	9.60	1
	Fatmire Sahiti	3932	1	9.40	2
	Pranati Rai Prakash	897	1	9.40	3

Q29. Get the following details for top 9 directors (based on number of movies):

```

QUERY- WITH top_directors AS(
SELECT  n.id as director_id, n.name as director_name,
        COUNT(m.id) AS movie_count,
        RANK() OVER (ORDER BY COUNT(m.id) DESC) as director_rank
FROM    names AS n INNER JOIN director_mapping AS d
        ON n.id=d.name_id INNER JOIN movie AS m
        ON d.movie_id = m.id
GROUP BY n.id),
movie_summary AS(
SELECT  n.id AS director_id, n.name AS director_name,
        m.id AS movie_id,  m.date_published, r.avg_rating,
        r.total_votes,  m.duration,
        LEAD(date_published) OVER (PARTITION BY n.id
        ORDER BY m.date_published) AS next_date_published,
        DATEDIFF(LEAD(date_published) OVER (PARTITION BY n.id
        ORDER BY m.date_published),date_published) AS inter_movie_days
FROM    names AS n INNER JOIN director_mapping AS d
        ON n.id=d.name_id INNER JOIN movie AS m
        ON d.movie_id = m.id INNER JOIN ratings AS r
        ON m.id=r.movie_id
WHERE   n.id IN (SELECT director_id
                FROM top_directors
                WHERE director_rank<=9))
SELECT  director_id, director_name,
        COUNT(DISTINCT movie_id) AS number_of_movies,
        ROUND(AVG(inter_movie_days),0) AS avg_inter_movie_days,
        ROUND(SUM(avg_rating*total_votes)/SUM(total_votes), 2) AS avg_rating,
        SUM(total_votes) AS total_votes, MIN(avg_rating) AS min_rating,
        MAX(avg_rating) AS max_rating,
        SUM(duration) AS total_duration
FROM    movie_summary
GROUP BY director_id
ORDER BY number_of_movies DESC, avg_rating DESC;
```

OUTPUT

director_id	director_name	number_of_movies	avg_inter_movie_days	avg_rating	total_votes	min_rating	max_rating	total_duration
nm1777967	A.L. Vijay	5	177	5.65	1754	3.7	6.9	611
nm2096009	Andrew Jones	5	191	3.04	1989	2.7	3.2	430
nm0001752	Steven Soderbergh	4	254	6.77	171684	6.2	7.0	401
nm0515005	Sam Liu	4	260	6.32	28557	5.8	6.7	311
nm0814469	Sion Sono	4	331	6.31	2972	5.4	6.4	500
nm0425364	Jesse V. Johnson	4	299	6.10	14778	4.2	6.5	380
nm2691863	Justin Price	4	315	4.93	5343	3.0	5.8	340
nm0831321	Chris Stokes	4	198	4.32	3664	4.0	4.6	350
nm6356309	Özgür Bakar	4	112	3.96	1092	3.1	4.9	374